TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 16. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: like terms and distribution

1. Combine like terms.

$$(x^2 + x - 1) + (-3x^2 - 2)$$

2. Combine like terms.

$$(-6x + 1) + (-2x^2 + 5x + 4)$$

3. Combine like terms.

$$(-3x^2 + 4x - 3) + (-2x^2 - 2x - 6)$$

4. Distribute and simplify.

$$2x(-x^2 - 3x + 3)$$

5. Distribute and simplify.

$$2x(-x^2 - 3x - 5)$$

6. Distribute and simplify.

$$2x(-x^2 - 3x - 4)$$

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Core: polynomial multiplication

7. Multiply and combine like terms.

$(x-5)(2x+3)$

8. Multiply and combine like terms.

$(x+4)(x+1)$

9. Multiply and combine like terms.

$(x-1)(2x+1)$

10. Multiply and combine like terms.

$(x+2)(2x-4)$

11. Multiply and combine like terms.

$(x^2+x+1)(x^2-x+3)$

12. Multiply and combine like terms.

$(x^2+x+2)(x^2-x+3)$

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Core: introductory factoring

13. Factor completely; check by multiplying.

$x^2 + 7x + 10$

14. Factor completely; check by multiplying.

$x^2 - 7x + 12$

15. Factor completely; check by multiplying.

$x^2 + 4x - 12$

16. Factor completely; check by multiplying.

$x^2 - 4x - 21$

17. Factor completely; check by multiplying.

$6x^2 + 12x$

18. Factor completely; check by multiplying.

$x^2 - 16$

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Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Extension: mixed polynomial practice

19. Multiply and simplify.

$$(x^2+2x+1)(x^2+x-2)$$

20. Multiply and simplify.

$$(2x^2+x-3)(x^2-x+2)$$

21. Multiply and simplify.

$$(x+2)(x^2-4)$$

22. Factor completely and verify.

$$x^2-3x-28$$

23. Factor completely and verify.

$$x^2-10x+25$$

24. Explain the error: only the first and last terms were multiplied.

$$(x+2)(x+3)=x^2+6$$

TSM 098 | ANSWER KEY

Polynomial operations and basic factoring

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: like terms and distribution

1. Combine like terms.

$$-2x^2 + x - 3$$

2. Combine like terms.

$$-2x^2 - x + 5$$

3. Combine like terms.

$$-5x^2 + 2x - 9$$

4. Distribute and simplify.

$$-2x^3 - 6x^2 + 6x$$

5. Distribute and simplify.

$$-2x^3 - 6x^2 - 10x$$

6. Distribute and simplify.

$$-2x^3 - 6x^2 - 8x$$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: polynomial multiplication

7. Multiply and combine like terms.

$2x^2 - 7x - 15$

8. Multiply and combine like terms.

$x^2 + 5x + 4$

9. Multiply and combine like terms.

$2x^2 - x - 1$

10. Multiply and combine like terms.

$2x^2 - 8$

11. Multiply and combine like terms.

$x^4 + 3x^2 + 2x + 3$

12. Multiply and combine like terms.

$x^4 + 4x^2 + x + 6$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Core: introductory factoring

13. Factor completely; check by multiplying.

$(x+2)(x+5)$

14. Factor completely; check by multiplying.

$(x-3)(x-4)$

15. Factor completely; check by multiplying.

$(x+6)(x-2)$

16. Factor completely; check by multiplying.

$(x-7)(x+3)$

17. Factor completely; check by multiplying.

$6x(x+2)$

18. Factor completely; check by multiplying.

$(x-4)(x+4)$

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Equivalent forms are acceptable. Use these answers after attempting the problems; revisit the linked lecture notes for additional examples.

Extension: mixed polynomial practice

19. Multiply and simplify.

$$x^4 + 3x^3 + x^2 - 3x - 2$$

20. Multiply and simplify.

$$2x^4 - x^3 + 5x - 6$$

21. Multiply and simplify.

$$x^3 + 2x^2 - 4x - 8$$

22. Factor completely and verify.

$$(x + 4)(x - 7)$$

23. Factor completely and verify.

$$(x - 5)(x - 5)$$

24. Explain the error: only the first and last terms were multiplied.

$$(x + 2)(x + 3) = x^2 + 5x + 6$$

Every term in one factor must multiply every term in the other. The missing middle products are 3x and 2x.